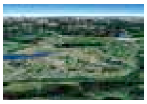


Particle accelerator bigger than LHC

Physists are planning a new particle accelerator with a 100km ring span which is nearly 4 times as long as LHC. **https://dgit.in/feb19-61**

Body heat to power smart garments

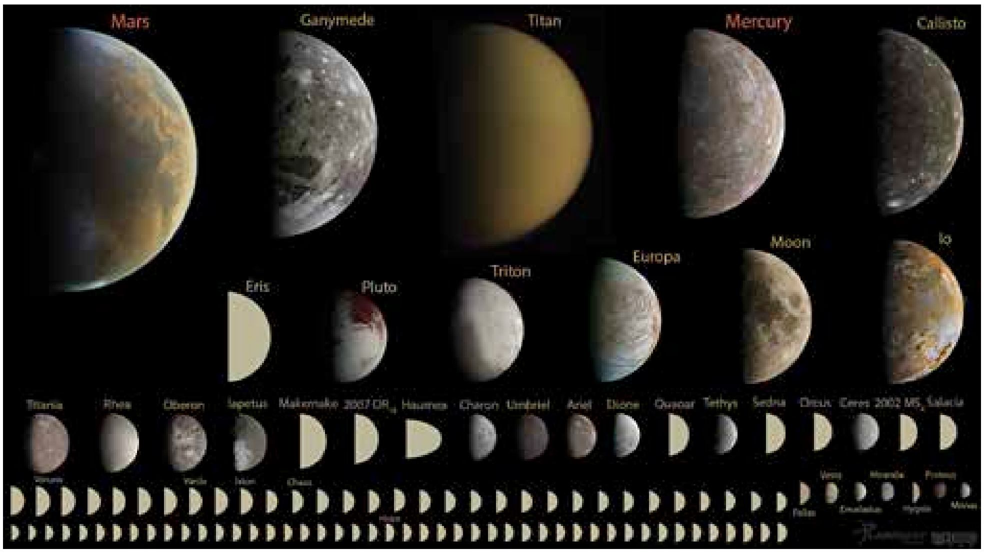
New thermoelectric fabric bands generate thermo-voltages greater than 20 milliVolts when worn on the hand. **https://dgit.in/feb19-62**

the eight objects considered to be planets by the IAU have actually cleared their orbits of other objects. The Earth would not be considered a planet because of the Near Earth Objects (NEOs), most of which are asteroids, but some of which are comets. One of these, designated as the asteroid 2016 HO3 is a quasi-satellite of the Earth, and is a specific example of Earth not having cleared its orbit. The asteroid accompanies the Earth in its annual journey around the Sun, but is too distant to be considered a true satellite. It disqualifies the Earth from being considered a planet, according to the definition of the IAU. The same criterion also makes it difficult to designate planets in other star systems. For example, for the 7 identified bodies in the Trappist-1 system to be called exoplanets, it would be necessary to find proof that they have in fact cleared their orbits of all other bodies. Considering how incredibly difficult it is to find exoplanets in the first place, this is a ridiculous exercise. Mars, Jupiter and Neptune also share their orbits with asteroids, technically disqualifying them from being considered planets according to the definition of the IAU.
The third criteria means that the farther away from the host star a body is, the more massive it has to be to be qualified as a planet. Even an Earth sized body as distant from the Sun as Pluto is, would not be able to entirely clear the orbit of all other bodies. Instead of defining a planet by characteristics intrinsic to the body in question, the IAU relies on external factors. The definition also introduces some inconsistencies. Mercury, Venus and Earth are referred to as terrestrial planets. Jupiter and Saturn as the Gas Giants. Neptune and Uranus are the Ice Giants. All of these bodies are considered planets, while only Pluto and the category of bodies called "dwarf planets", are not considered planets. Pluto satisfies the first and second criteria of being a planet, but not the third. It is therefore purely arbitrary that Pluto is not considered a planet. According to planetary scientists, the IAU is saying "Pluto is not a planet because we say it is not a planet", and not providing sufficient and robust scientific reasons, even in the definition that they have adopted.

Planetary scientists have proposed a definition based on the geophysics of the body, one that is simple enough to understand, and one that is in line with how people actually use the word. The proposed definition is: A planet is a sub-stellar mass body that has never undergone nuclear fusion and that has sufficient self-gravitation to assume a spheroidal shape adequately described by a triaxial ellipsoid regardless of its orbital parameters. A simpler definition for school textbooks, would be: "Round objects in space that are smaller than Stars".

It was not just the planetary scientists that have a problem with the demotion of Pluto. The new definition of the IAU ended up confusing the general public as well. The New Horizons spacecraft was a probe launched to study objects in the outer reaches in the Solar System, and Pluto was one of the bodies that it studied. One of the most common questions asked to the New Horizons team, was why was a probe being sent to study Pluto, when it was no longer a planet. Being removed from the roster of planets in orbit around the Sun, somehow made Pluto an object less interesting for study in the minds of the people. The science team members of the New Horizons mission are leading the effort to get Pluto reclassified as a planet.

The proposed definition by the planetary scientists is dependent on the internal characteristics of the body, and not the external environment. If the new definition is adopted, the number of planets in the Solar System would expand significantly. Instead of having 8 planets, the Solar System would have more than 110 planets. This is because apart from the asteroids and stars, every other body is considered to be a planet. This includes the moons of Jupiter and Saturn, as well as our own Moon. Ganymede, a moon of Jupiter, and Titan, a moon of Saturn are both larger than Mercury. This categorisation actually helps give the public a clearer understanding of the nature of these worlds. The definition is more useful than the IAU version to planetary scientists who study the geology and and other geosciences of these worlds.



The planets in the Solar System, to scale, according to the definition based on geophysics.